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Studierichting: Informatica
Oefeningenreeks 4A nr. 30.10

$$\int \frac{2x^3dx}{\sqrt{x^2+4}}$$

$$u = x^2 + 4$$

$$du = 2xdx$$

$$2x^3dx = x^2(2xdx)$$

$$x^2 = u - 4$$

$$\Rightarrow \int \frac{x^2(2xdx)}{\sqrt{x^2+4}}$$

$$= \int \frac{u-4}{\sqrt{u}}du$$

$$= \int \frac{u}{\sqrt{u}}du - 4\int \frac{1}{\sqrt{u}}du$$

$$= \int u^{1-\frac{1}{2}}du - 4\int u^{-1/2}du$$

$$= \frac{2}{3}(x^2+4)^{\frac{3}{2}} - 8(x^2+4)^{\frac{1}{2}} + C$$

$$= \frac{2}{3}\sqrt{(x^2+4)^3} - 8\sqrt{x^2+4} + C$$

References